

Advanced Econometrics: Exercise #8

September 22, 2025

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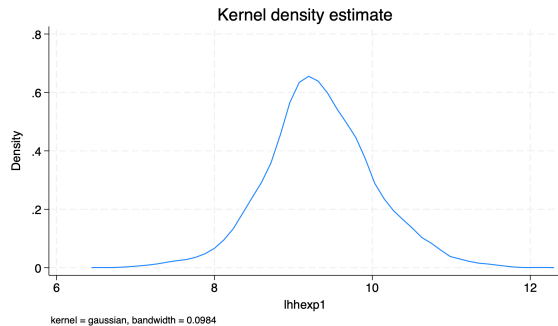
Amy Qian

Problem 1

Obtain a non-parametric estimate of the density of `lhhexp1`. Compare the bandwidth used by STATA with the one suggested by Silverman's rule of thumb.

STATA

```
kdensity lhhexp1, kernel(gaussian)
```



Bandwidth used by STATA: 0.0984

Silverman's Rule of Thumb

```
sum lhhexp1, d
```

This gives us

- Standard deviation: 0.6877458
- n: 5,999
- IQR: 9.759566 - 8.919547 = 0.840019

To calculate Silverman's bandwidth, we use the formula: $h = \min(\sigma, \frac{IQR}{1.34}) \frac{0.9}{n^{-1/5}}$

Plugging in our values, we get:

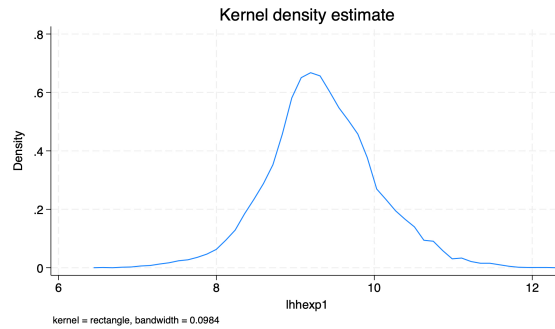
- `display ((.6877458*0.9)/5999^(0.2)) = .10865624`
- `display (((9.759566 - 8.919547)/1.34*0.9)/5999^(0.2)) = .09904009`

So the bandwidth suggested by Silverman's rule of thumb is 0.09904009, which is very close to the bandwidth used by STATA.

Problem 2

Repeat the previous exercise using different kernels.

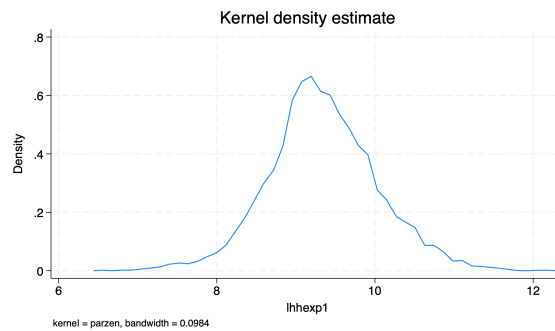
Rectangular Kernel



`kdensity lhexp1, kernel(rec)`

This is less smooth than the Gaussian kernel.

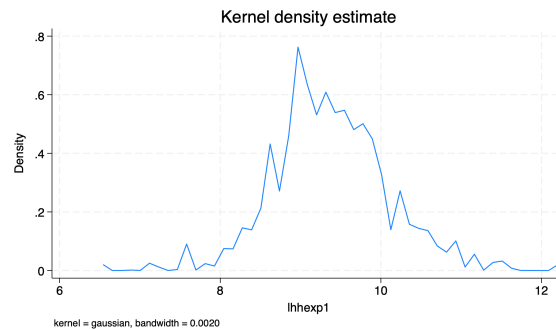
Parzan Kernel



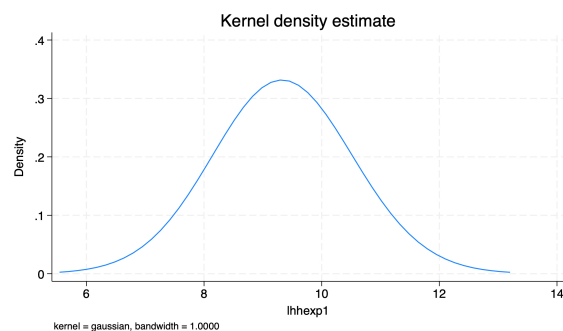
`kdensity lhexp1, kernel(parz)`

This is also less smooth than the Gaussian kernel.

Gaussian with different bandwidth



`kdensity lhexp1, kernel(gaussian) bwidth(0.002)`



`kdensity lhexp1, kernel(gaussian) bwidth(1)`

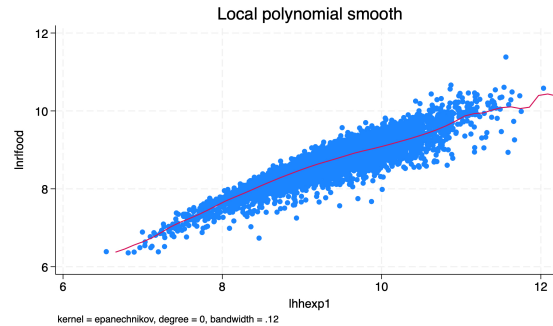
In general, the bandwidth is more important than the kernel. We want it to be smooth enough but not too smooth that we miss important features of the data.

Problem 3

Obtain the Nadaraya-Watson estimator of the conditional expectation of `lnrlfood`, the log household expenditure on food, given `lhhexp1`. Does the result suggest that this conditional expectation is linear?

Solution

We want to estimate $E[y_i|x_i] = m(x_i)$ where y_i is `lnrlfood`.



`lpoly lnrlfood lhhexp1`

Blue dots are the data points, the red line is the estimated conditional expectation (the function). Could be linear looking at the graph, but not exactly linear.

Problem 4

Using OLS, estimate a regression of `lnrlfood` on `lhhexp1`, `age` and its square. Is the effect of age linear?

Solution

First generate the new variable age squared.

`gen age2 = age^2`

Then run the OLS regression using robust standard errors to control for heteroskedasticity.

`reg lnrlfood lhhexp1 age age2, r`

We want to test the null hypothesis that the effect of age is linear, which is equivalent to testing if the coefficient of `age2` is zero.

$$H_0 : \beta_{\text{age2}} = 0 \text{ vs. } H_a : \beta_{\text{age2}} \neq 0$$

From the regression output, we see that the p-value is 0.000, which is less than 0.05, so we reject H_0 in favor of H_1 at 5% confidence level.

Problem 5

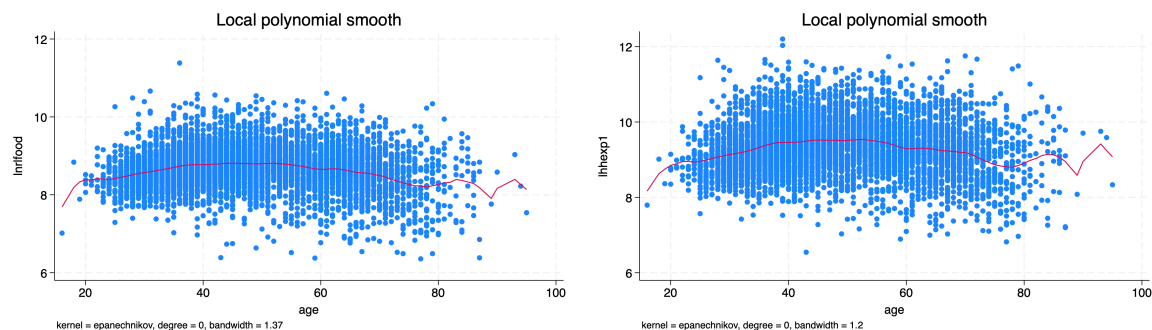
Use the partially linear estimator to estimate in the following model: $\text{lnrlfood}_i = \beta \text{lhhexp1}_i + g(\text{age}_i) + u_i$

Solution

$G(\cdot)$ is now an unknown function.

Get the estimates

- $E[y_i|\text{age}_i] \Rightarrow \text{foodage}$. `lpoly lnrlfood age, generate(foodage) at (age)`
- $E[x_i|\text{age}_i] \Rightarrow \text{expage}$. `lpoly lhhexp1 age, generate(expage) at (age)`



Now we compute the differences.

- $\text{gen } y = \text{lnrlfood} - \text{foodage}$
- $\text{gen } x = \text{lhhexp1} - \text{expage}$

Finally, we run the regression of y on x , not assuming anything on the distribution of the data.

```

Source |      SS      df      MS      Number of obs      =
-----+-----+-----+-----+-----+-----
> 5,999 |              |              |              |              |
> 64.09 |              |              |              |              |
> .0000 | Model      | 1274.81681    1 1274.81681    Prob > F      = 0
> .0000 | Residual   | 324.491644   5,998 .054099974    R-squared     = 0
> .7971 |              |              |              |              |
> .7971 |              |              |              |              |
> .7971 | Total      | 1599.30845   5,999 .266595841    Adj R-squared = 0
> 23259 |              |              |              |              |
> Root MSE = .266595841

-----+-----+-----+-----+-----+-----
> y | Coefficient Std. err.      t    P>|t|     [95% conf. inte
> rval]
-----+-----+-----+-----+-----+-----
> x |      .695743   .0045324   153.51   0.000     .686858     .70
> 46281 |

```

reg y x, noconstant

Problem 6

Compare the results.

```

Linear regression               Number of obs      =      5,999
                                F(3, 5995)          =     5484.44
                                Prob > F            =     0.0000
                                R-squared            =     0.8112
                                Root MSE          =     .2333

```

		Robust				
lnrlfood	Coefficient	std. err.	t	P> t	[95% conf. interval]	
lhhexp1	.6942858	.005711	121.57	0.000	.6830901	.7054815
age	.005095	.0016213	3.14	0.002	.0019168	.0082733
age2	-.0000716	.0000161	-4.45	0.000	-.0001031	-.00004
_cons	2.1277	.0561216	37.91	0.000	2.017681	2.237718

Actually quite close.